

Experiment 'p_4_6_find_minimum_array' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_6_find_minimum_array

Natural language statement: null

Method signature: p_4_6_find_minimum_array(inputs: array<int>) returns (min: int)

Ensures

- forall i :: 0 <= i < inputs.Length ==> min <= inputs[i]
- exists i :: 0 <= i < inputs.Length && min == inputs[i]

Requires

- 0 < inputs.Length

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_6_find_minimum_array(inputs: array<int>) returns (min: int)`

The method should respect the following contract:

requires `0 < inputs.Length`, ensures forall `i :: 0 <= i < inputs.Length ==> min <= inputs[i]`
↪ `i]`, ensures exists `i :: 0 <= i < inputs.Length && min == inputs[i]`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnMEU4XDheo3TJfVAVMcEjKv8LOom

Model: gpt-5.5-2026-04-23

Created at: 1780655906

Finish reason: stop

Usage: 316 tokens in, and 566 tokens out

```
{
  min := inputs[0];
  var idx := 1;
  while idx < inputs.Length
    invariant 1 <= idx <= inputs.Length
    invariant forall i :: 0 <= i < idx ==> min <= inputs[i]
    invariant exists i :: 0 <= i < idx && min == inputs[i]
  {
    if inputs[idx] < min {
      min := inputs[idx];
    }
    idx := idx + 1;
  }
}
```

Final Program

```
method p_4_6_find_minimum_array(inputs: array<int>) returns (min: int)
```

```

    requires 0 < inputs.Length
    ensures forall i :: 0 <= i < inputs.Length ==> min <= inputs[i]
    ensures exists i :: 0 <= i < inputs.Length && min == inputs[i]
{
  min := inputs[0];
  var idx := 1;
  while idx < inputs.Length
    invariant 1 <= idx <= inputs.Length
    invariant forall i :: 0 <= i < idx ==> min <= inputs[i]
    invariant exists i :: 0 <= i < idx && min == inputs[i]
  {
    if inputs[idx] < min {
      min := inputs[idx];
    }
    idx := idx + 1;
  }
}

```

Total Token Usage

Input tokens: 316

Output tokens: 566

Reasoning tokens: 441

Sum of ‘total tokens’: 882

Experiment Timings

Overall Experiment started at 1780655906131, ended at 1780655935184, lasting 29053ms (29.05 seconds)

Iteration #1 started at 1780655906133, ended at 1780655935184, lasting 29051ms (29.05 seconds)